JUNIOR SCHOOL LEARNING HUB

GRADE 8 MATHEMATICS
DETAILED LEARNER NOTES – 2027

School: _____
Learner: _____
Grade: 8 | Learning Area: Mathematics | Year: 2027

HOW TO USE THESE NOTES

- Read the explanation and learn the mathematical vocabulary before attempting the examples.
- Study each worked example line by line. Copy the method and explain why each step was taken.
- Attempt the practice exercises without looking at the answers. Show all working and include units.
- Use a ruler, compass, graph paper or calculator only where the task requires it.
- For construction and graph questions, submit clear drawings with labels and suitable scales.
- The notes are arranged by the Grade 8 Mathematics strands and sub-strands in the supplied 2027 schemes of work.

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TERM I – 1. NUMBERS

Integers

An integer is a whole number that may be positive, negative or zero. Integers do not include fractions or decimals. Positive integers are greater than zero; negative integers are less than zero.

Detailed Explanation and Key Concepts

• Integers are used to describe temperatures above or below zero, deposits and withdrawals, gains and losses, and floors above or below ground level.

• On a number line, zero is the reference point. Positive integers are placed to the right of zero and negative integers to the left.

• The farther an integer is to the right, the greater its value. For example, –2 is greater than –5.

• For addition, move right for a positive number and left for a negative number. For subtraction, add the opposite of the number being subtracted.

• For combined operations, use brackets first, then multiplication or division, and finally addition or subtraction from left to right.

• Comparing integers: imagine walking along a straight path marked with numbers. The number farther to the right is always greater. Thus 2 > –1 and –3 > –8. The negative sign tells us direction or position, not that the number is automatically smaller in magnitude.

• Opposites: two integers are opposites if they are the same distance from zero but on opposite sides. The opposite of +6 is –6. Their sum is zero.

• Addition signs: when the signs are the same, add the absolute values and keep the common sign. When signs differ, subtract the smaller absolute value from the larger and use the sign of the number with the larger absolute value.

• Subtraction rule: change subtraction into addition of the opposite. For example, 7 – (–4) becomes 7 + 4, while –3 – 5 becomes –3 + (–5).

• Multiplication and division: equal signs give a positive answer; different signs give a negative answer. Multiplication and division follow the same sign rule.

Worked Examples

Example 1: Example 1: Represent –3, +2 and 0 on a number line.

Working and explanation: Mark zero at the centre. Place –3 three equal units to the left, +2 two units to the right, and 0 at the origin.

Example 2: Example 2: Calculate –4 + 7.

Working and explanation: Start at –4 and move 7 units to the right. The result is 3.

Example 3: Example 3: Calculate 5 – (–3).

Working and explanation: Subtracting a negative is equivalent to adding its opposite: 5 + 3 = 8.

Example 4: Example: Evaluate –8 + 3 – (–5).

Working and explanation: Change subtraction of a negative to addition: –8 + 3 + 5. First –8 + 3 = –5; then –5 + 5 = 0.

Example 5: Example: A diver is 6 m below sea level and rises 9 m. Represent the final position.

Working and explanation: Below sea level is –6 m. Rising 9 m gives –6 + 9 = +3 m. The diver is 3 m above sea level.

Example 6: Example: Evaluate (–4) × (–6) ÷ 3.

Working and explanation: A negative multiplied by a negative is positive: 24 ÷ 3 = 8.

Practice Exercise

1. Classify –8, 0, 12, 3.5 and ¼ as integers or non-integers.

2. Arrange –6, 4, –1, 0 and 3 in ascending order.

3. Evaluate: –7 + 12; 8 – 13; –5 – (–9); 6 + (–3) × 2.

4. A lift is on floor –2 and rises 7 floors. State its new floor.

5. Evaluate: (–12) + 7; 15 – 23; –9 – (–4); (–6) × 7; 48 ÷ (–8).

6. Arrange –11, 6, –2, 0, 9 and –7 in descending order.

7. A bank account has a balance of KSh –350. A deposit of KSh 500 is made. Find the new balance.

8. At 6 a.m. the temperature was –4°C. It rose by 11°C and later fell by 6°C. Find the final temperature.

9. Evaluate 18 – [–3 × (4 – 7)].

10. Explain why –5 is greater than –9 using a number line.

Learning Resources and Practical Activities

Number line; integer cards; thermometer; floor-level diagrams; calculator or digital integer game.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM I – 1. NUMBERS

Fractions

A fraction represents part of a whole or a quotient. In a/b, a is the numerator and b is the denominator, where b is not zero.

Detailed Explanation and Key Concepts

• Equivalent fractions have the same value. Multiply or divide the numerator and denominator by the same non-zero number to obtain an equivalent fraction.

• Simplify a fraction by dividing the numerator and denominator by their highest common factor.

• For addition and subtraction, first express fractions using a common denominator. Add or subtract the numerators and retain the denominator.

• For multiplication, multiply numerators together and denominators together. For division, multiply by the reciprocal of the second fraction.

• In combined operations, evaluate brackets first, multiplication and division next, then addition and subtraction from left to right.

• A word problem can be solved by identifying the whole quantity, the fraction involved and the operation required.

• Proper fractions have a numerator smaller than the denominator; improper fractions have a numerator equal to or greater than the denominator. A mixed number contains a whole number and a proper fraction.

• To convert a mixed number to an improper fraction, multiply the whole number by the denominator, add the numerator, and place the result over the original denominator. For example, 2 1/3 = (2×3+1)/3 = 7/3.

• To compare fractions, use a common denominator or convert them to decimals. For positive fractions with the same denominator, the fraction with the larger numerator is greater.

• Cancellation can make multiplication easier. Divide a numerator and a denominator by a common factor before multiplying, but only cancel factors, not terms joined by addition or subtraction.

• Word-problem strategy: underline the quantities, identify what the question asks, decide whether to add, subtract, multiply or divide, show the calculation, and state the answer with its unit.

Worked Examples

Example 1: Example 1: Add 2/3 + 1/4.

Working and explanation: The lowest common denominator is 12. Therefore 2/3 = 8/12 and 1/4 = 3/12. Sum = 11/12.

Example 2: Example 2: Multiply 3/5 × 10/9.

Working and explanation: Multiply: 30/45. Simplify by 15 to obtain 2/3.

Example 3: Example 3: Divide 4/7 by 2/3.

Working and explanation: 4/7 × 3/2 = 12/14 = 6/7.

Example 4: Example 4: A family used 2/5 of a bag of flour in the morning and 1/4 in the evening. What fraction was used?

Working and explanation: 2/5 + 1/4 = 8/20 + 5/20 = 13/20.

Example 5: Example: Convert 3 2/5 to an improper fraction.

Working and explanation: 3 × 5 + 2 = 17, so 3 2/5 = 17/5.

Example 6: Example: Subtract 2 1/4 – 3/8.

Working and explanation: Convert 2 1/4 to 2 2/8. Then 2 2/8 – 3/8 = 1 7/8.

Example 7: Example: A recipe uses 3/4 kg of flour per cake. How much is needed for 4 cakes?

Working and explanation: 4 × 3/4 = 12/4 = 3 kg.

Practice Exercise

1. Work out: 5/6 – 1/3; 3/4 + 2/5; 7/8 × 4/7; 5/6 ÷ 10/9.

2. Evaluate: 1/2 + 3/4 × 2/3.

3. A tank was 3/4 full. If 1/3 of its capacity was used, what fraction remained?

4. A learner spent 2/5 of pocket money on lunch and 1/4 on transport. What fraction remained?

5. Convert 4 3/7 and 2 5/6 into improper fractions.

6. Simplify 42/56 and 63/81.

7. Work out 5/8 + 7/12.

8. Work out 3 1/2 – 1 3/4.

9. Calculate 2/3 × 9/10 ÷ 3/5.

10. A water tank was 5/6 full. If 1/4 of its capacity was drained, what fraction remained?

11. A trader sold 2/5 of a sack of maize in the morning and 1/3 in the afternoon. What fraction was left?

12. Create a word problem involving division of fractions and solve it.

Learning Resources and Practical Activities

Fraction strips; paper folding; shopping price cards; exercise book; calculator where appropriate.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM I – 1. NUMBERS

Decimals

Decimals are numbers written using a decimal point to separate the whole-number part from the fractional part. Each place to the right of the decimal point represents tenths, hundredths, thousandths and so on.

Detailed Explanation and Key Concepts

• Convert a fraction to a decimal by dividing the numerator by the denominator. Some fractions give terminating decimals while others give recurring decimals.

• A recurring decimal has one or more digits that repeat indefinitely. A recurring digit or group is indicated using a bar, for example 0.333… = 0.3■.

• To round to a given number of decimal places, identify the required last digit and inspect the digit immediately after it. Increase the last retained digit by one if the next digit is 5 or more.

• Significant figures begin at the first non-zero digit. Retain the required number of significant digits and use the next digit to decide whether to round up.

• Standard form is written as A × 10■, where 1 ≤ A < 10 and n is an integer.

• For addition and subtraction, align decimal points. For multiplication, multiply normally and place the decimal point using the total number of decimal places. For division, make the divisor a whole number by shifting both decimal points equally.

• Place value: in 38.472, 3 represents 30, 8 represents 8, 4 represents four tenths, 7 represents seven hundredths and 2 represents two thousandths.

• Terminating decimals end after a finite number of digits, such as 0.25. Recurring decimals continue with a repeating pattern, such as 0.666… or 0.272727….

• To convert a recurring decimal to a fraction, let the decimal equal x, multiply by a power of 10 that moves the repeating block, subtract the original equation, and solve.

• Rounding is an estimate, not an exact calculation. The place value required must be stated clearly. Do not confuse decimal places with significant figures.

• Standard form is useful for very large and very small numbers. A positive power of ten is used for large numbers; a negative power is used for small numbers.

Worked Examples

Example 1: Example 1: Convert 3/8 to a decimal.

Working and explanation: 3 ÷ 8 = 0.375.

Example 2: Example 2: Round 18.476 to two decimal places.

Working and explanation: The third decimal digit is 6, so 18.47 rounds to 18.48.

Example 3: Example 3: Write 5 600 000 in standard form.

Working and explanation: 5 600 000 = 5.6 × 10■.

Example 4: Example 4: Calculate 4.25 + 0.78.

Working and explanation: Align decimal points: 4.25 + 0.78 = 5.03.

Example 5: Example: Convert 0.375 to a fraction in simplest form.

Working and explanation: 0.375 = 375/1000. Divide numerator and denominator by 125: 3/8.

Example 6: Example: Round 0.004786 to 2 significant figures.

Working and explanation: The first two significant digits are 4 and 7. The next digit is 8, so round 7 up. Answer = 0.0048.

Example 7: Example: Calculate 7.2 ÷ 0.3.

Working and explanation: Multiply both numbers by 10: 72 ÷ 3 = 24.

Practice Exercise

1. Convert 7/20 and 5/6 to decimals.

2. Round 43.7864 to two decimal places and to three significant figures.

3. Express 0.000072 in standard form.

4. Calculate: 12.6 – 4.875; 2.4 × 0.35; 8.64 ÷ 0.12.

5. Write 0.625 as a fraction in simplest form.

6. Convert 7/16 to a decimal.

7. Round 56.7894 to 3 decimal places.

8. Round 0.006784 to 2 significant figures.

9. Express 72 500 000 and 0.0000084 in standard form.

10. Calculate 14.08 + 3.976 – 5.8.

11. Calculate 3.75 × 0.24.

12. Calculate 18.9 ÷ 0.07.

13. A learner buys 2.5 kg of rice at KSh 180 per kg. Find the cost.

Learning Resources and Practical Activities

Place-value chart; decimal cards; calculators; shopping receipts; number line.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM I – 1. NUMBERS

Squares and Square Roots

The square of a number is the number multiplied by itself. The square root of a non-negative number is a value which, when multiplied by itself, gives the original number.

Detailed Explanation and Key Concepts

• If $n \times n = n^2$, then $\sqrt{(n^2)} = n$ for $n \geq 0$. Examples of perfect squares include 1, 4, 9, 16, 25, 36, 49, 64, 81 and 100.

• A square table gives the squares of numbers. A square-root table can be used to read or estimate square roots.

• For numbers above 100, separate the number into a convenient form or use tables as directed. Always check the answer by squaring it.

• The square of a decimal less than 1 is smaller than the original positive decimal. For example, $0.4^2 = 0.16$.

• A calculator may be used to find squares and square roots. Enter the number carefully and use the square-root key where appropriate.

• Square numbers form a sequence: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196 and 225. Learning these helps with quick estimation.

• Square roots undo squaring. Since $12^2 = 144$, $\sqrt{144} = 12$. In this level, the principal square root symbol $\sqrt{}$ denotes the non-negative root.

• To estimate a non-perfect square, find the two consecutive perfect squares between which it lies. For example, 30 lies between 25 and 36, so $\sqrt{30}$ lies between 5 and 6.

• When using mathematical tables, identify the correct row, column and any mean difference required by the table layout. Follow the table instructions and retain the required accuracy.

• Check a square-root answer by multiplying it by itself. A small rounding difference may occur when the answer is given to a fixed number of decimal places.

Worked Examples

Example 1: Example 1: Find 17^2.

Working and explanation: $17 \times 17 = 289$.

Example 2: Example 2: Find $\sqrt{196}$.

Working and explanation: Since $14 \times 14 = 196$, $\sqrt{196} = 14$.

Example 3: Example 3: Find 0.6^2.

Working and explanation: $0.6 \times 0.6 = 0.36$.

Example 4: Example 4: Estimate $\sqrt{50}$.

Working and explanation: $7^2 = 49$ and $8^2 = 64$, so $\sqrt{50}$ is slightly greater than 7.

Example 5: Example: Find 23^2 using $(20 + 3)^2$.

Working and explanation: $(20 + 3)^2 = 20^2 + 2 \times 20 \times 3 + 3^2 = 400 + 120 + 9 = 529$.

Example 6: Example: Estimate $\sqrt{90}$ to the nearest whole number.

Working and explanation: $9^2 = 81$ and $10^2 = 100$. Since 90 is closer to 81 than 100, $\sqrt{90}$ is approximately 9 to the nearest whole number.

Example 7: Example: Find the square root of 0.0049.

Working and explanation: $0.07 \times 0.07 = 0.0049$, so $\sqrt{0.0049} = 0.07$.

Practice Exercise

1. Find 13^2, 25^2 and 0.8^2.

2. Find $\sqrt{144}$, $\sqrt{225}$ and $\sqrt{400}$.

3. Between which two whole numbers does $\sqrt{70}$ lie?

4. Use a calculator to find $\sqrt{72}$ to two decimal places and verify by squaring.

5. Complete the squares: 12^2, 15^2, 18^2, 21^2 and 30^2.

6. Find $\sqrt{169}$, $\sqrt{256}$, $\sqrt{324}$ and $\sqrt{625}$.

7. Estimate $\sqrt{45}$ and $\sqrt{110}$ between consecutive whole numbers.

8. Find the square of 0.25 and 1.2.

9. Use a calculator to find $\sqrt{85}$ to 2 decimal places.

10. A square garden has area 196 m^2. Find its side length and perimeter.

11. Explain why 50 is not a perfect square.

Learning Resources and Practical Activities

Square-number chart; mathematical tables; calculator; square grid.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM I – 2. NUMBERS

Rates, Ratios, Proportions and Percentages

A rate compares quantities measured in different units, such as kilometres per hour or shillings per kilogram. A ratio compares quantities of the same kind or quantities expressed in compatible units.

Detailed Explanation and Key Concepts

• A rate is calculated by dividing one quantity by another. Include the correct units in the answer.

• A ratio a:b compares a parts of one quantity to b parts of another. Simplify a ratio by dividing all terms by their highest common factor.

• To compare ratios, express them in equivalent forms or divide each term by the same quantity to obtain a unit rate.

• To divide a quantity in a given ratio, add the ratio parts, divide the total quantity by the sum of the parts, then multiply by each part.

• Percentage increase = (increase ÷ original value) × 100%. Percentage decrease = (decrease ÷ original value) × 100%.

• In direct proportion, when one quantity increases or decreases, the other changes in the same ratio. The quotient y/x remains constant.

• In indirect (inverse) proportion, one quantity increases as the other decreases so that their product remains constant, in the stated situation.

Worked Examples

Example 1: Example 1: A car travels 180 km in 3 hours. Find its average speed.

Working and explanation: Speed = distance ÷ time = 180 ÷ 3 = 60 km/h.

Example 2: Example 2: Divide KSh 840 in the ratio 3:4.

Working and explanation: Total parts = 7. One part = 840 ÷ 7 = 120. Shares are 3 × 120 = KSh 360 and 4 × 120 = KSh 480.

Example 3: Example 3: Increase 500 by 12%.

Working and explanation: 12% of 500 = 60. New value = 500 + 60 = 560.

Example 4: Example 4: Four workers complete a task in 6 days. At the same rate, how long would 8 workers take?

Working and explanation: Workers × days is constant: 4 × 6 = 24 worker-days. 24 ÷ 8 = 3 days.

Practice Exercise

1. Find the rate in 240 km travelled in 4 hours.

2. Simplify 18:24 and divide 1 200 in the ratio 2:3.

3. A price rises from KSh 800 to KSh 920. Find the percentage increase.

4. State whether each is direct or inverse proportion: number of items and total cost at a fixed price; number of workers and time for a fixed task.

Learning Resources and Practical Activities

Ratio cards; measuring tools; price lists; hourglass; containers; calculators.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM I – 3. ALGEBRA

Algebraic Expressions

An algebraic expression is a mathematical phrase containing numbers, variables and operation signs. A term is a part of an expression separated by plus or minus signs. A factor is a quantity multiplied by another quantity.

Detailed Explanation and Key Concepts

• Like terms have the same variables raised to the same powers. Unlike terms do not. Only like terms can be combined by addition or subtraction.

• The coefficient is the numerical factor of a term. In 7x, the coefficient is 7.

• Factorisation is the reverse of expansion. Identify the highest common factor (HCF) of all terms, then write the expression as a product.

• An algebraic fraction is a fraction containing algebraic expressions. Simplify by factorising the numerator and denominator and cancelling common non-zero factors.

• A denominator must not be zero. State or observe restrictions where relevant.

• Digital algebra tools may be used to check expansions and factorisations, but learners should show the mathematical steps.

• Collecting like terms: 3x + 5x = 8x, but 3x + 5y cannot be simplified to 8xy or 8x. The variable part must be identical.

• Expansion uses the distributive law: a(b + c) = ab + ac. For example, 3(x + 4) = 3x + 12.

• Factorisation reverses expansion. Find the greatest common factor shared by every term, place it outside brackets and divide each term by it.

• Algebraic fractions must be simplified by factorising first. A common factor can be cancelled only when it is a factor of the whole numerator and denominator.

• Substitution means replacing a variable with a known value. Use brackets when substituting a negative number, for example x^2 when x = –3 gives $(–3)^2 = 9$.

Worked Examples

Example 1: Example 1: Simplify 4x + 3x – 2.

Working and explanation: Combine like terms: 7x – 2.

Example 2: Example 2: Factorise 6x + 9.

Working and explanation: The HCF is 3. Therefore 6x + 9 = 3(2x + 3).

Example 3: Example 3: Simplify $(x^2 – 9)/(x + 3)$.

Working and explanation: Factorise the numerator: $(x – 3)(x + 3)/(x + 3) = x – 3$, provided $x \neq –3$.

Example 4: Example 4: Identify like terms in 5a, 3b, –2a and 7ab.

Working and explanation: 5a and –2a are like terms. 3b and 7ab are unlike terms.

Example 5: Example: Expand 4(2x – 3).

Working and explanation: Multiply every term inside the bracket by 4: 8x – 12.

Example 6: Example: Factorise $12a^2b + 18ab^2$.

Working and explanation: The HCF is 6ab. Therefore $12a^2b + 18ab^2 = 6ab(2a + 3b)$.

Example 7: Example: Simplify $(6x^2y)/(9xy^2)$.

Working and explanation: Divide coefficients by 3, cancel x and one y: 2x/(3y), where x and y are non-zero.

Practice Exercise

1. Simplify 8x – 3x + 4 – 9.

2. Factorise 10y + 15.

3. Simplify $(x^2 + 5x)/x$, stating the restriction.

4. Explain why $4p$ and $4p^2$ are unlike terms.

5. Simplify $7a + 4b - 3a + 6b$.

6. Expand $5(3x - 2)$.

7. Factorise $15x + 25$.

8. Factorise $8a^2 - 12a$.

9. Simplify $(x^2 - 16)/(x + 4)$, stating the restriction.

10. Simplify $12m^2n \div 4mn$.

11. Evaluate $3x^2 - 2x$ when $x = -2$.

12. Explain the difference between expansion and factorisation.

Learning Resources and Practical Activities

Algebra tiles; expression cards; charts; exercise book; ICT algebra tools.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM I – 3. ALGEBRA

Linear Equations in Two Unknowns

A linear equation in two unknowns contains two variables, usually x and y, each with a power of one. A pair of simultaneous linear equations has a common solution that satisfies both equations.

Detailed Explanation and Key Concepts

• Translate words into algebraic expressions by identifying unknown quantities and the relationships between them.

• In the substitution method, make one variable the subject in one equation, substitute that expression into the other equation, solve, then substitute back.

• In the elimination method, multiply one or both equations where necessary so that the coefficients of one variable are equal or opposites. Add or subtract the equations to eliminate that variable.

• Check a solution by substituting both values into the original equations.

• Simultaneous equations can represent real-life situations involving prices, quantities, ages, lengths and costs.

• An equation is balanced: whatever operation is performed on one side must also be performed on the other side. This preserves equality.

• Forming equations: choose symbols for unknowns, translate each statement into an equation, and check that the equations represent the situation correctly.

• Substitution works well when one variable is already the subject or can easily be made the subject. Keep each algebraic step on a separate line.

• Elimination works well when coefficients are equal or can easily be made equal. Align like terms vertically before adding or subtracting.

• Some pairs of equations may have one solution, no solution or infinitely many solutions. In Grade 8 practical work, focus on finding and checking the common solution represented by the given pair.

Worked Examples

Example 1: Example 1: Solve y = x + 2 and x + y = 8.

Working and explanation: Substitute y = x + 2: x + x + 2 = 8; 2x = 6; x = 3. Then y = 5.

Example 2: Example 2: Solve x + y = 9 and x – y = 3.

Working and explanation: Add the equations: 2x = 12, so x = 6. Substitute into x + y = 9: y = 3.

Example 3: Example 3: A pen and a book cost KSh 70. Two pens and one book cost KSh 100. Form equations.

Working and explanation: Let p be the pen price and b the book price. p + b = 70; 2p + b = 100. Subtracting gives p = 30, then b = 40.

Example 4: Example: Solve 2x + y = 13 and x + y = 8 by elimination.

Working and explanation: Subtract the second equation from the first: x = 5. Substitute into x + y = 8: y = 3. Solution (5,3).

Example 5: Example: Solve x + y = 12 and y = 2x.

Working and explanation: Substitute: x + 2x = 12; 3x = 12; x = 4. Then y = 8.

Example 6: Example: Three notebooks and two pens cost KSh 160. One notebook and two pens cost KSh 80. Find each price.

Working and explanation: Let n be notebook price and p pen price. 3n + 2p = 160; n + 2p = 80. Subtract: 2n = 80, so n = 40. Then 40 + 2p = 80, so p = 20.

Practice Exercise

1. Solve by substitution: y = 2x + 1 and x + y = 10.

2. Solve by elimination: 2x + y = 11 and x + y = 7.

3. Two adult tickets and one child ticket cost KSh 550. One adult ticket and one child ticket cost KSh 350. Find each ticket price.

4. Check each solution in both original equations.

5. Solve by substitution: $y = 3x - 2$ and $x + y = 10$.

6. Solve by elimination: $3x + y = 14$ and $x + y = 8$.

7. Two adult and three child tickets cost KSh 1 100. One adult and one child ticket cost KSh 500. Find each price.

8. Form two equations for two numbers whose sum is 25 and difference is 7, then solve.

9. Check the solution $x = 4$, $y = 3$ in $2x + y = 11$ and $x - y = 1$.

10. Explain one advantage of elimination and one advantage of substitution.

Learning Resources and Practical Activities

Equation balance diagrams; shopping scenario cards; graph paper; calculators.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM II – 1. MEASUREMENTS

Circles: Circumference, Arc Length and Sector Perimeter

A circle is a plane shape whose points are all the same distance from its centre. The radius is the distance from the centre to the circumference. The diameter passes through the centre and is twice the radius.

Detailed Explanation and Key Concepts

• Circumference is the distance around a circle. $C = \pi d$ or $C = 2\pi r$, where π is approximately 22/7 or 3.142.

• Measure the circumference of a circular object by wrapping a string around it, marking the length and measuring the string with a ruler.

• An arc is part of a circle's circumference. The length of an arc is proportional to its central angle.

• Arc length = $(\theta/360°) \times 2\pi r$, where θ is the angle at the centre.

• A sector is the region enclosed by two radii and an arc. Its perimeter is the arc length plus the two radii: P = arc length + 2r.

• The radius joins the centre to the circumference. The diameter passes through the centre and joins two points on the circumference. Therefore $d = 2r$ and $r = d/2$.

• Pi (π) is the constant ratio of circumference to diameter for every circle. It is approximately 3.142. Use 22/7 when instructed or when it makes the calculation convenient.

• Always identify whether the question gives radius or diameter before substituting into the formula. If diameter is given, use $C = \pi d$; if radius is given, use $C = 2\pi r$.

• An angle at the centre determines the fraction of the full circumference represented by an arc. A 90° arc is one quarter of the circumference because 90/360 = 1/4.

• The perimeter of a sector includes the curved arc and both straight radii. Do not report only the arc length when the question asks for sector perimeter.

Worked Examples

Example 1: Example 1: Find the circumference of a circle of radius 7 cm.

Working and explanation: $C = 2\pi r = 2 \times 22/7 \times 7 = 44$ cm.

Example 2: Example 2: Find the arc length for r = 14 cm and θ = 90°.

Working and explanation: Arc = $90/360 \times 2 \times 22/7 \times 14 = 22$ cm.

Example 3: Example 3: Find the perimeter of the sector in Example 2.

Working and explanation: Perimeter = 22 + 14 + 14 = 50 cm.

Example 4: Example: A wheel has diameter 70 cm. Find the distance travelled in one complete revolution.

Working and explanation: Distance in one revolution = circumference = $\pi d = 22/7 \times 70 = 220$ cm.

Example 5: Example: Find the arc length for a 120° sector of radius 21 cm.

Working and explanation: Arc = $120/360 \times 2 \times 22/7 \times 21 = 1/3 \times 132 = 44$ cm.

Example 6: Example: Find the perimeter of a 60° sector of radius 14 cm.

Working and explanation: Arc = $60/360 \times 2 \times 22/7 \times 14 = 44/6 = 22/3$ cm. Perimeter = 14 + 14 + 22/3 = 106/3 = 35 1/3 cm.

Practice Exercise

1. Find the circumference of a circle with diameter 21 cm.

2. A wheel has radius 35 cm. Find its circumference.

3. Find the arc length when r = 10 cm and θ = 72° using π = 3.142.

4. Find the perimeter of a sector with radius 7 cm and arc length 11 cm.

5. Find the circumference of a circle of radius 10.5 cm using 22/7.

6. A circular track has diameter 140 m. Find one lap distance.

7. Find the arc length of a 45° sector with radius 28 cm.

8. Find the perimeter of a 120° sector of radius 7 cm.

9. A wheel of circumference 2.2 m makes 500 revolutions. Find the distance travelled in metres and kilometres.

10. Explain the difference between radius, diameter and circumference.

Learning Resources and Practical Activities

String; ruler; circular objects; bicycle wheel; compass; calculator.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM II – 1. MEASUREMENTS

Area and Surface Area

Area measures the surface covered by a two-dimensional shape. Surface area is the total area of the outside surfaces of a three-dimensional solid.

Detailed Explanation and Key Concepts

• Area of a circle: $A = \pi r^2$. Area of a sector: $A = (\theta/360°) \times \pi r^2$.

• A cube has six equal square faces. Its surface area is $6a^2$, where a is the length of an edge.

• A cuboid has three pairs of equal rectangular faces. Its surface area is $2(lw + lh + wh)$.

• A cylinder has two circular ends and one curved surface. Curved surface area = $2\pi rh$. Total surface area of a closed cylinder = $2\pi r^2 + 2\pi rh$. For an open cylinder, exclude the missing circular end.

• A triangular prism has two triangular ends and three rectangular faces. Surface area is the sum of the areas of all five faces.

• For irregular shapes on a square grid, count complete squares and estimate partial squares using a consistent method. State the unit squared.

• Map area estimates require attention to the map scale; convert lengths before calculating area.

• Area is measured in square units such as cm^2 or m^2. Convert all lengths to a consistent unit before using a formula.

• For a sector, the fraction $\theta/360°$ tells us what fraction of the full circle is included. Multiply this fraction by the area of the whole circle.

• A net helps prevent missing a face. Label each face, calculate its area and add all external faces. Do not include faces that are not part of the outside surface.

• For a triangular prism, calculate the areas of the two triangular ends and the three rectangular faces. The total is the sum of all five faces.

• For irregular shapes on a grid, count complete squares and combine partial squares sensibly. An estimate should be identified as an estimate.

Worked Examples

Example 1: Example 1: Find the area of a circle of radius 7 cm.

Working and explanation: $A = \pi r^2 = 22/7 \times 7 \times 7 = 154\ cm^2$.

Example 2: Example 2: Find the surface area of a cube of side 4 cm.

Working and explanation: $SA = 6a^2 = 6 \times 16 = 96\ cm^2$.

Example 3: Example 3: Find the surface area of a cuboid 5 cm by 3 cm by 2 cm.

Working and explanation: $SA = 2(5\times3 + 5\times2 + 3\times2) = 2(15 + 10 + 6) = 62\ cm^2$.

Example 4: Example 4: Find the curved surface area of a cylinder with r = 3 cm and h = 8 cm.

Working and explanation: $CSA = 2\pi rh = 2 \times \pi \times 3 \times 8 = 48\pi\ cm^2$.

Example 5: Example: Find the area of a 90° sector of radius 14 cm.

Working and explanation: $Area = 90/360 \times 22/7 \times 14^2 = 1/4 \times 616 = 154\ cm^2$.

Example 6: Example: Find the total surface area of a closed cylinder with radius 7 cm and height 10 cm.

Working and explanation: $TSA = 2\pi r^2 + 2\pi rh = 2\times22/7\times49 + 2\times22/7\times7\times10 = 308 + 440 = 748\ cm^2$.

Example 7: Example: A triangular prism has a triangular end of area 12 cm^2 and three rectangular faces of areas 20 cm^2, 24 cm^2 and 30 cm^2. Find its surface area.

Working and explanation: $Surface\ area = 2\times12 + 20 + 24 + 30 = 98\ cm^2$.

Practice Exercise

1. Find the area of a circle of diameter 14 cm.

2. Find the area of a 60° sector of a circle with radius 6 cm.

3. Calculate the surface area of a cube of side 8 cm and a cuboid 8 cm × 5 cm × 3 cm.

4. Explain the difference between the surface area of an open and a closed cylinder.

5. Find the area of a circle of radius 10.5 cm.

6. Find the area of a 72° sector with radius 10 cm.

7. Find the surface area of a cube with edge 9 cm.

8. Find the surface area of a cuboid 12 cm × 7 cm × 4 cm.

9. Find the curved surface area of a cylinder with radius 5 cm and height 12 cm.

10. A triangular prism has two triangular faces each of area 15 cm² and three rectangular faces of areas 30 cm², 40 cm² and 50 cm². Find total surface area.

11. Estimate the area of an irregular shape drawn on a 1 cm square grid and explain your method.

Learning Resources and Practical Activities

Grid paper; nets; cardboard models; circular objects; rulers; calculators.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM II – 2. MONEY

Interest, Appreciation, Depreciation and Hire Purchase

Principal is the original amount of money borrowed or invested. Interest is the additional money paid or earned. The rate is usually expressed as a percentage per annum.

Detailed Explanation and Key Concepts

• Simple interest is calculated on the original principal only: I = (P × R × T)/100, where P is principal, R is annual rate (%) and T is time in years.

• Amount for simple interest = P + I.

• Compound interest is calculated on the principal and accumulated interest. Amount A = P(1 + r/100)■ for annual compounding, where n is the number of years. Compound interest = A – P.

• Appreciation is an increase in value over time. For annual appreciation at a fixed rate, new value = previous value × (1 + rate/100).

• Depreciation is a decrease in value over time. New value = previous value × (1 – rate/100).

• Hire purchase allows a buyer to pay a deposit followed by instalments. Total hire-purchase price = deposit + total instalments. Compare it with the cash price to find the extra cost.

• When using financial institutions or digital shopping platforms, compare charges, interest, terms and security of transactions.

• Interest is the cost of borrowing or the reward for saving/investing. Read the question carefully to identify whether it asks for interest only or the final amount.

• Time must be expressed in years when the annual rate is used in the simple-interest formula. For example, 6 months = 6/12 = 0.5 years.

• Compound interest differs from simple interest because each period's interest is added to the balance before the next period is calculated.

• Appreciation and depreciation are repeated percentage changes. For more than one year, apply the percentage to the value at the beginning of each year, not always to the original value.

• With hire purchase, total instalments means the amount of one instalment multiplied by the number of instalments. Add the deposit once only.

Worked Examples

Example 1: Example 1: Find simple interest on KSh 10 000 at 8% p.a. for 3 years.

Working and explanation: I = 10 000 × 8 × 3 ÷ 100 = KSh 2 400. Amount = KSh 12 400.

Example 2: Example 2: Find the amount on KSh 5 000 at 10% compound interest for 2 years.

Working and explanation: $A = 5\,000(1.10)^2 = 5\,000 \times 1.21 =$ KSh 6 050. Compound interest = KSh 1 050.

Example 3: Example 3: A machine worth KSh 40 000 depreciates by 10% in one year.

Working and explanation: New value = 40 000 × 0.90 = KSh 36 000.

Example 4: Example 4: A product has a deposit of KSh 2 000 and 10 monthly instalments of KSh 900.

Working and explanation: Hire-purchase price = 2 000 + 10 × 900 = KSh 11 000.

Example 5: Example: Find simple interest on KSh 12 000 at 9% per annum for 8 months.

Working and explanation: Time = 8/12 year. I = 12 000 × 9 × 8/12 ÷ 100 = KSh 720.

Example 6: Example: Find the compound amount on KSh 10 000 at 10% per annum for 3 years.

Working and explanation: $A = 10\,000 \times 1.1^3 = 10\,000 \times 1.331 =$ KSh 13 310.

Example 7: Example: A computer worth KSh 50 000 depreciates by 20% per year for two years.

Working and explanation: After year 1: 50 000 × 0.8 = 40 000. After year 2: 40 000 × 0.8 = KSh 32 000.

Practice Exercise

1. Calculate simple interest on KSh 15 000 at 6% p.a. for 4 years.

2. Find the compound amount on KSh 8 000 at 5% p.a. for 2 years.

3. A vehicle worth KSh 600 000 appreciates by 4% in one year. Find its new value.

4. A phone costs KSh 18 000 cash or a deposit of KSh 4 000 plus 12 instalments of KSh 1 400. Find the hire-purchase price and extra cost.

5. Find simple interest on KSh 25 000 at 7% p.a. for 2 years.

6. Find the amount on KSh 18 000 at 5% simple interest for 9 months.

7. Calculate compound amount on KSh 20 000 at 10% p.a. for 2 years.

8. A machine worth KSh 80 000 depreciates by 15% in one year. Find its value after one year.

9. A house worth KSh 2 000 000 appreciates by 5% per annum for one year. Find its new value.

10. A fridge costs KSh 35 000 cash or KSh 5 000 deposit plus 15 monthly instalments of KSh 2 200. Find total hire-purchase price and extra cost.

11. Explain two differences between simple and compound interest.

Learning Resources and Practical Activities

Price lists; loan and savings examples; calculators; sample receipts; secure digital shopping demonstrations.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM II – 3. GEOMETRY

Geometrical Constructions and Polygons

Geometrical construction is the accurate drawing of lines, angles and shapes using instruments such as a ruler, compass and set square.

Detailed Explanation and Key Concepts

• Parallel lines remain the same distance apart and never meet. They can be constructed using a compass and ruler or a set square and ruler.

• Perpendicular lines meet at 90°. A perpendicular bisector crosses a line segment at its midpoint and at right angles.

• A perpendicular from a point to a line is a line segment meeting the given line at 90°.

• Proportional division divides a line segment into parts in a given ratio.

• The sum of interior angles of an n-sided polygon is (n – 2) × 180°.

• The sum of exterior angles of any convex polygon, taking one at each vertex in the same direction, is 360°.

• A regular polygon has equal sides and equal interior angles. A regular triangle is an equilateral triangle; a regular quadrilateral is a square.

• Use a sharp pencil and keep the compass point steady. Construction arcs should remain visible until the required line has been drawn and checked.

• To construct parallel lines with a ruler and compass, use equal perpendicular distances or transfer an angle accurately. With a set square and ruler, slide the set square along the ruler while keeping its edge fixed.

• To construct a perpendicular bisector, draw arcs of equal radius from both endpoints of a line segment, using a radius greater than half the segment. Join the two arc intersections.

• To construct a perpendicular from a point on a line, mark equal distances on either side of the point and use intersecting arcs to locate a second point on the perpendicular.

• To divide a line proportionally, draw an auxiliary ray from one endpoint, mark equal intervals on it, join the last mark to the other endpoint and draw parallel lines through the intermediate marks.

• A polygon is a closed plane figure made of straight line segments. A regular polygon has all sides and all interior angles equal.

• Interior-angle sum is (n–2)×180°. For a regular polygon, divide the sum by n to find one interior angle. Each exterior angle of a regular polygon is 360°/n.

• To construct regular polygons, use a compass to set equal side lengths and a protractor or geometric construction to set equal angles. Check side lengths and angles after drawing.

Worked Examples

Example 1: Example 1: Find the sum of interior angles of a hexagon.

Working and explanation: (6 – 2) × 180° = 720°.

Example 2: Example 2: Find each interior angle of a regular pentagon.

Working and explanation: Sum = (5 – 2) × 180° = 540°. Each angle = 540° ÷ 5 = 108°.

Example 3: Construction reminder: Perpendicular bisector.

Working and explanation: Open the compass to more than half the line length. Draw arcs from each endpoint above and below the line. Join the arc intersections with a ruler.

Example 4: Example: Find the interior-angle sum of a decagon.

Working and explanation: n = 10. Sum = (10 – 2) × 180° = 1 440°.

Example 5: Example: Find each exterior angle of a regular 15-sided polygon.

Working and explanation: Each exterior angle = 360° ÷ 15 = 24°.

Example 6: Example: A polygon has an interior-angle sum of 900°. How many sides does it have?

Working and explanation: (n – 2) × 180 = 900. Divide by 180: n – 2 = 5, so n = 7 sides.

Practice Exercise

1. Find the sum of interior angles of an octagon.
2. Find each interior angle of a regular hexagon.
3. Explain how to construct a perpendicular bisector.
4. Construct a regular triangle, square and regular pentagon using suitable instruments.
5. Construct a perpendicular bisector of a 7 cm line segment and show the construction arcs.
6. Construct a perpendicular from a point outside a given line.
7. Construct two parallel lines 4 cm apart using a ruler and set square.
8. Find the interior-angle sum of a 12-sided polygon.
9. Find each interior angle of a regular octagon.
10. A regular polygon has an exterior angle of 30°. Find its number of sides.
11. Construct an equilateral triangle and a square using ruler and compass.
12. Explain how to divide a line segment into three equal parts.

Learning Resources and Practical Activities

Ruler; compass; set squares; protractor; sharp pencil; plain paper.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM II – 4. ALGEBRA

Coordinates and Graphs

The Cartesian plane is formed by two perpendicular number lines: the horizontal x-axis and the vertical y-axis. Their intersection is the origin (0,0). A point is represented by an ordered pair (x,y).

Detailed Explanation and Key Concepts

• The x-coordinate is read first and the y-coordinate second. The plane is divided into four quadrants.

• Choose a suitable and uniform scale on each axis. Label axes and include units where necessary.

• To plot a point, move horizontally from the origin according to x, then vertically according to y.

• For a linear equation, make a table of values by substituting selected x-values and calculating the corresponding y-values.

• Plot the ordered pairs accurately and join them with a straight line where the relationship is linear.

• The point where two straight-line graphs intersect represents the solution of the corresponding simultaneous equations.

• A graphical solution can be checked by substituting the coordinates into both equations.

• Use a consistent scale on each axis. The horizontal axis is labelled x and the vertical axis y. The axes meet at the origin, (0,0).

• Quadrants are numbered anticlockwise from the top-right: Quadrant I has (+,+), II has (–,+), III has (–,–), and IV has (+,–). Points on an axis are not in a quadrant.

• When completing a table of values, substitute each chosen x-value into the equation carefully. Use brackets for negative values.

• A straight-line graph should have accurately plotted points, a ruler-drawn line, a suitable scale, labelled axes and a clear title where appropriate.

• To solve simultaneous equations graphically, draw both lines on the same axes. Read the coordinates of their intersection and check the values in both equations.

Worked Examples

Example 1: Example 1: Plot (–2,3).

Working and explanation: From the origin, move 2 units left and 3 units up.

Example 2: Example 2: Complete a table for y = 2x + 1 when x = 0, 1 and 2.

Working and explanation: When x = 0, y = 1; when x = 1, y = 3; when x = 2, y = 5. Points: (0,1), (1,3), (2,5).

Example 3: Example 3: Solve y = x + 1 and y = 5 – x graphically.

Working and explanation: The lines intersect when x + 1 = 5 – x, giving x = 2 and y = 3. The graphical solution is (2,3).

Example 4: Example: Complete the table for y = 2x – 3 when x = –1, 0, 1, 2.

Working and explanation: Substitute: x=–1 gives y=–5; x=0 gives –3; x=1 gives –1; x=2 gives 1. Points: (–1,–5), (0,–3), (1,–1), (2,1).

Example 5: Example: Determine the intersection of y = 2x and y = 6 – x.

Working and explanation: At intersection, 2x = 6 – x. Thus 3x = 6, x = 2 and y = 4. The lines meet at (2,4).

Practice Exercise

1. State the quadrant of each point: (3,4), (–2,5), (–3,–1), (4,–2).

2. Make a table for y = 3x – 2 when x = –1, 0, 1 and 2.

3. Draw the graph of y = 2x + 1 using a suitable scale.

4. Find the intersection of y = x + 2 and y = 6 – x using a table and graph.

5. Plot A(2,3), B(–3,2), C(–2,–4) and D(4,–1). State each quadrant.

6. Complete a table for y = 3x + 2 when x = –2, –1, 0, 1 and 2.

7. Draw the graph $y = 2x - 1$ using a suitable scale.

8. Read the coordinates of three points from a graph you have drawn.

9. Draw $y = x + 2$ and $y = 8 - 2x$ on the same axes and estimate their intersection.

10. Explain why the axes must have a consistent scale.

Learning Resources and Practical Activities

Graph paper; ruler; pencils; Cartesian plane charts; calculators; graphing software.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM III – 4. GEOMETRY

Scale Drawing

A scale drawing is a representation of an actual object or distance using a fixed ratio. The scale compares a length on the drawing or map with the corresponding actual length.

Detailed Explanation and Key Concepts

• Always express both lengths in the same units before forming or using a scale.

• Scale length = actual length ÷ scale factor when the scale is expressed as drawing:actual in ratio form 1:n. Actual length = scale length × n.

• A statement scale may read, for example, 1 cm represents 2 km. A ratio scale may be written 1:200 000 after converting both lengths to the same unit.

• To convert statement scale to ratio form, convert the actual distance to the same unit as the drawing length, then write the ratio and simplify.

• To convert ratio scale to statement form, interpret the ratio using a convenient drawing unit and convert the actual length to a suitable unit.

• Use a ruler and the stated scale to calculate map distances and prepare accurate scale drawings.

• In application problems, identify the scale, convert units, calculate the required length and label the drawing clearly. ICT mapping tools can help measure and compare distances.

• A scale is a fixed ratio between a drawing length and the corresponding actual length. A scale drawing keeps the shape and proportions of the object but changes its size.

• Unit conversion is essential. For example, 1 km = 100 000 cm. Thus a scale of 1:100 000 means 1 cm on the map represents 1 km on the ground.

• For a ratio scale 1:n, drawing length = actual length ÷ n and actual length = drawing length × n, provided both lengths use the same unit.

• A statement scale such as 1 cm represents 4 km can be used directly when the map distance is measured in centimetres and the actual distance is required in kilometres.

• Choose a scale that allows the drawing to fit the available paper while remaining easy to measure. State the scale clearly and label all dimensions.

Worked Examples

Example 1: Example 1: A map scale is 1 cm represents 5 km. What actual distance is represented by 4 cm?

Working and explanation: 4 × 5 = 20 km.

Example 2: Example 2: Convert 1:100 000 to a statement scale.

Working and explanation: 1 cm represents 100 000 cm = 1 km.

Example 3: Example 3: Actual length is 12 m and scale is 1:200. Find drawing length.

Working and explanation: 12 m = 1 200 cm. Drawing length = 1 200 ÷ 200 = 6 cm.

Example 4: Example: A map scale is 1:50 000. Two towns are 6 cm apart on the map. Find the actual distance.

Working and explanation: Actual distance = 6 × 50 000 = 300 000 cm. Convert: 300 000 ÷ 100 000 = 3 km.

Example 5: Example: A room is 5 m long. Draw it using a scale of 1:100.

Working and explanation: Convert 5 m to 500 cm. Drawing length = 500 ÷ 100 = 5 cm.

Example 6: Example: A road is 12 km long and is represented by 4 cm. Write the scale as a statement and ratio.

Working and explanation: 4 cm represents 12 km, so 1 cm represents 3 km. Since 3 km = 300 000 cm, ratio scale = 1:300 000.

Practice Exercise

1. A map uses 1 cm to represent 3 km. Find the actual distance for 7.5 cm.

2. Convert 1:50 000 into a statement scale.

3. A wall is 8 m long. Find its drawing length at scale 1:100.

4. Explain why actual lengths must be converted to the same units before calculating a scale.

5. A map scale is 1:25 000. Find the actual distance represented by 8 cm.

6. A distance of 15 km is represented by 5 cm. Write the scale in statement and ratio form.

7. A rectangular field is 60 m by 40 m. Draw it using a scale of 1:1 000.

8. A room measures 4.8 m by 3.6 m. Find its dimensions on a plan drawn to scale 1:50.

9. Explain why scale drawings are useful to architects and engineers.

10. Convert a scale of 1 cm to 250 m into ratio form.

Learning Resources and Practical Activities

Maps; floor plans; ruler; tape measure; pencils; calculators; digital map tools.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM III – 4. GEOMETRY

Common Solids, Nets and Surface Area

Common solids are three-dimensional shapes. They occupy space and have measurable dimensions. A face is a flat surface, an edge is where two faces meet, and a vertex is a corner.

Detailed Explanation and Key Concepts

• A cube has 6 square faces, 12 edges and 8 vertices. A cuboid has 6 rectangular faces, 12 edges and 8 vertices.

• A cylinder has two circular flat faces and one curved surface. A cone has one circular base and one curved surface meeting at a vertex.

• A prism has two equal, parallel end faces and rectangular side faces. A pyramid has a base and triangular faces meeting at an apex.

• A net is a two-dimensional arrangement of faces that can be folded to form a solid. Check that faces join correctly without overlap.

• Surface area is found by adding the areas of all external faces. For a cube, SA = $6a^2$. For a cuboid, SA = 2(lw + lh + wh).

• For a cylinder, total surface area of a closed cylinder = $2\pi r^2 + 2\pi rh$. For an open cylinder, omit the area of the open end.

• A cone's curved surface is a sector when opened into a net. Its curved surface area is πrl, where l is the slant height; total surface area adds πr^2 for the base.

• Models can be made from card or other safe materials. ICT drawing tools can support accurate nets and solid representations.

• Count faces, edges and vertices carefully. A face is a flat surface, an edge is a line where two faces meet, and a vertex is a point where edges meet.

• A cube net consists of six equal squares arranged so that they can fold into a cube. Not every arrangement of six squares is a valid cube net.

• A cuboid net consists of six rectangles arranged in three matching pairs. Opposite faces are equal.

• A cylinder net consists of one rectangle and two equal circles. The rectangle's length equals the circumference of the circular base and its width equals the cylinder height.

• A cone net consists of one circular base and a sector of a larger circle. The sector's radius is the cone's slant height.

• For a pyramid, the net contains the base and triangular faces. The surface area is the base area plus the areas of all triangular faces.

• Distance along a surface can be found by opening the solid into a net and drawing the shortest straight path on the flat net, then measuring or calculating that path.

Worked Examples

Example 1: Example 1: Find the surface area of a cube with edge 5 cm.

Working and explanation: SA = $6 \times 5^2 = 150$ cm².

Example 2: Example 2: Find the surface area of a cuboid 6 cm × 4 cm × 3 cm.

Working and explanation: SA = 2(24 + 18 + 12) = 108 cm².

Example 3: Example 3: Find the total surface area of a closed cylinder with r = 3 cm and h = 7 cm.

Working and explanation: TSA = $2\pi r^2 + 2\pi rh = 18\pi + 42\pi = 60\pi$ cm².

Example 4: Example 4: A solid has 5 faces, 9 edges and 6 vertices. Identify it if it has triangular ends.

Working and explanation: It is a triangular prism.

Example 5: Example: A cuboid has length 8 cm, width 5 cm and height 3 cm. Find its total surface area.

Working and explanation: TSA = 2(lw + lh + wh) = 2(40 + 24 + 15) = 158 cm².

Example 6: Example: A cylinder has radius 4 cm and height 9 cm. Find its curved surface area.

Working and explanation: CSA = $2\pi rh = 2 \times \pi \times 4 \times 9 = 72\pi$ cm².

Example 7: Example: A cone has radius 3 cm and slant height 5 cm. Find its total surface area.

Working and explanation: TSA = $\pi rl + \pi r^2 = \pi \times 3 \times 5 + \pi \times 9 = 15\pi + 9\pi = 24\pi$ cm².

Practice Exercise

1. State the number of faces, edges and vertices of a cube and cuboid.
2. Sketch a possible net of a cube and a cylinder.
3. Find the surface area of a cuboid measuring 10 cm × 4 cm × 2 cm.
4. Explain the difference between curved surface area and total surface area.
5. Draw and label a net of a cube.
6. Draw a net of a cylinder and label the rectangle dimensions in terms of r and h.
7. State the faces, edges and vertices of a triangular prism.
8. Find the total surface area of a cube of edge 6 cm.
9. Find the total surface area of a cylinder of radius 3.5 cm and height 10 cm.
10. A cone has radius 7 cm and slant height 12 cm. Find its total surface area.
11. Describe how a net can be used to find the shortest distance along the surface of a solid.

Learning Resources and Practical Activities

Packaging boxes; nets; cardboard; scissors; ruler; glue; calculators; 3D drawing software.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM III – 5. STATISTICS

Data Presentation and Interpretation

Data are facts or observations collected for a purpose. Discrete data are countable values. Data can be organised in tables and displayed using bar graphs and line graphs.

Detailed Explanation and Key Concepts

• Collect data using a clear question, observation or simple survey. Record responses accurately and organise them in a frequency table.

• A bar graph uses separate bars to represent categories or discrete values. Bars should have equal widths and spaces between them. Label axes and use a suitable scale.

• A line graph joins plotted points and is useful for showing change, especially over time.

• To interpret a graph, read the scale carefully, compare values, identify maximum and minimum values, and describe trends or differences.

• Mode is the value occurring most frequently. Mean = sum of values ÷ number of values. Median is the middle value after arranging data in order; for an even number of values, it is the mean of the two middle values.

• ICT tools can organise data, calculate averages and create graphs. Check that the entered data and graph labels are correct.

• Before drawing a graph, decide what each axis represents, identify the range of values and select a scale that uses the graph space well.

• A tally chart records frequencies using tally marks. Group tally marks in fives to make counting easier, then transfer the totals to a frequency table.

• A bar graph is suitable for comparing categories. Bars are separate because categories are distinct. A line graph is useful for showing a trend or change over time.

• Mean uses every value and can be affected by very large or very small values. Median is the middle value and is less affected by extreme values. Mode is the most frequent value.

• For an even number of observations, arrange the data and find the mean of the two middle values to obtain the median.

• Interpretation should use evidence from the graph: quote values, compare categories and describe increases, decreases or patterns rather than making unsupported statements.

Worked Examples

Example 1: Example 1: Find the mean of 4, 6, 8 and 10.

Working and explanation: Mean = (4 + 6 + 8 + 10) ÷ 4 = 28 ÷ 4 = 7.

Example 2: Example 2: Find the median of 3, 8, 5, 9 and 4.

Working and explanation: Arrange: 3, 4, 5, 8, 9. The middle value is 5.

Example 3: Example 3: Find the mode of 2, 3, 3, 4, 5, 3, 6.

Working and explanation: The value 3 occurs most often, so mode = 3.

Example 4: Example: Find the mean and median of 5, 7, 4, 9, 10.

Working and explanation: Mean = 35/5 = 7. Ordered data: 4,5,7,9,10. Median = 7.

Example 5: Example: Find the median of 3, 5, 6, 8, 10, 12.

Working and explanation: There are 6 values. Median = (6 + 8)/2 = 7.

Example 6: Example: A bar graph shows 12 learners chose football, 8 chose netball and 5 chose athletics. How many learners were surveyed?

Working and explanation: Total = 12 + 8 + 5 = 25 learners.

Practice Exercise

1. Find the mean, median and mode of 2, 4, 4, 6, 9.

2. Collect data on the number of books read by ten learners and present it in a bar graph.

3. Draw a line graph to show a temperature recorded at different times of the day.

4. State two conclusions that can be made from a graph showing weekly attendance.

5. The scores are 6, 7, 8, 8, 9, 10, 8, 6. Find mean, median and mode.

6. Draw a bar graph for the data: apples 12, oranges 8, bananas 15, mangoes 5.

7. Record the temperature at five different times and draw a line graph.

8. State two differences between a bar graph and a line graph.

9. Explain how to choose a suitable scale for a graph.

10. Give one advantage and one limitation of the mean, median and mode.

Learning Resources and Practical Activities

Survey sheets; tally charts; graph paper; rulers; calculators; spreadsheet software.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

TERM III – 5. STATISTICS

Probability

Probability measures how likely an event is to happen. It ranges from 0 to 1, where 0 represents an impossible event and 1 represents a certain event.

Detailed Explanation and Key Concepts

• An experiment is an activity with an uncertain outcome, such as tossing a coin or rolling a die.

• An outcome is a possible result. An event is one or more outcomes of interest.

• An event may be impossible, unlikely, equally likely, likely or certain.

• Experimental probability = number of times the event occurs ÷ total number of trials.

• For equally likely outcomes, theoretical probability = number of favourable outcomes ÷ total number of possible outcomes.

• Probability may be expressed as a fraction, decimal or percentage. Multiply a decimal by 100 to convert it to a percentage.

• The more trials carried out, the more stable experimental results may become, although they may not exactly equal theoretical probability.

• Digital probability games can simulate many trials and help compare experimental and theoretical results.

• An event is a set of one or more outcomes. For a fair coin, the sample space is {H,T}; for a fair six-sided die it is {1,2,3,4,5,6}.

• Complementary events are an event and its opposite. Their probabilities add to 1. For example, P(not getting a six) = 1 – P(getting a six) = 5/6.

• For equally likely outcomes, count the favourable outcomes and divide by the total number of possible outcomes. Make sure each outcome is equally likely before using this method.

• Experimental probability is based on results observed in trials. It may differ from theoretical probability, especially when only a small number of trials is carried out.

• To express a probability as a decimal, divide the numerator by the denominator. To express it as a percentage, multiply the decimal by 100%.

Worked Examples

Example 1: Example 1: A fair coin is tossed. Find the probability of getting a head.

Working and explanation: There are 2 equally likely outcomes and 1 favourable outcome. P(head) = 1/2.

Example 2: Example 2: A die is rolled. Find the probability of getting an even number.

Working and explanation: Even outcomes are 2, 4 and 6. P(even) = 3/6 = 1/2.

Example 3: Example 3: A coin is tossed 40 times and heads appears 23 times. Find experimental probability of a head.

Working and explanation: 23/40 = 0.575 = 57.5%.

Example 4: Example: A bag contains 4 red and 6 blue counters. Find P(red).

Working and explanation: Total counters = 10. P(red) = 4/10 = 2/5.

Example 5: Example: Find the probability of not getting an even number when a fair die is rolled.

Working and explanation: Odd outcomes are 1,3,5. P(odd) = 3/6 = 1/2. This is the complement of getting an even number.

Example 6: Example: A spinner lands on green 18 times in 60 spins. Find the experimental probability.

Working and explanation: P(green) = 18/60 = 3/10 = 0.3 = 30%.

Practice Exercise

1. A bag contains 3 red, 2 blue and 5 green counters. Find the probability of selecting a blue counter.

2. State whether each event is impossible, unlikely, likely or certain: rolling a 7 on a normal die; getting a number less than 7; getting a head on a fair coin.

3. A die is rolled 60 times and a six appears 8 times. Find the experimental probability of a six.

4. Express 3/8 as a decimal and percentage.

5. A bag contains 5 yellow, 3 red and 2 blue balls. Find P(yellow), P(red) and P(not blue).

6. A fair die is rolled. Find the probability of getting a number greater than 4.

7. A coin is tossed 100 times and tails occurs 54 times. Find experimental probability of tails.

8. Write 0.35 as a fraction and percentage.

9. A spinner has 8 equal sectors: 3 red, 2 blue and 3 green. Find P(blue) and P(not red).

10. Explain why experimental probability may differ from theoretical probability.

Learning Resources and Practical Activities

Coins; dice; counters; spinners; tally sheets; calculators; digital probability games.

Practical task: Work with a partner or a small group to demonstrate one concept using available materials. Record your procedure, calculations and conclusion.

Self-assessment: Can I explain the key terms, select the correct rule or formula, show accurate working and apply the concept to a real-life situation?

PRACTICE EXERCISE – SUGGESTED ANSWERS

Use these answers to mark your work after attempting the exercises. For construction and graph tasks, compare accuracy, labels, scale and method with teacher guidance.

Integers

1. –5; –8; –5; –42; –6.

2. 9, 6, 0, –2, –7, –11.

3. KSh 150.

4. 1°C.

5. 18 – [–3 × (–3)] = 18 – 9 = 9.

6. On a number line –5 lies to the right of –9.

Fractions

1. 31/7; 17/6.

2. 3/4; 7/9.

3. 29/24 = 1 5/24.

4. 1 3/4.

5. 1.

6. 7/12.

7. 4/15.

8. Answers will vary; check correct operation and working.

Decimals

1. 5/8.

2. 0.4375.

3. 56.789.

4. 0.0068.

5. 7.25×10■; 8.4×10■■.

6. 12.256.

7. 0.9.

8. 270.

9. KSh 450.

Squares and Square Roots

1. 144, 225, 324, 441, 900.

2. 13, 16, 18, 25.

3. 6<√45<7; 10<√110<11.

4. 0.0625; 1.44.

5. 9.22 approximately.

6. 14 m; 56 m.

7. No integer multiplied by itself gives 50.

Rates, Ratio, Proportions and Percentages

1. 30 km/h.

2. 1:4.

3. KSh 1 500 and KSh 900.

4. 7 litres.

5. 20%.

6. KSh 1 700.

7. 9 days.

8. 4.

Algebraic Expressions

1. 4a + 10b.

2. 15x – 10.

3. 5(3x + 5).

4. 4a(2a – 3).

5. x – 4, x ≠ –4.

6. 3m.

7. 16.

8. Expansion removes brackets by multiplying; factorisation writes terms as a product.

Linear Equations in Two Unknowns

1. x=3, y=7.

2. x=3, y=5.

3. Adult KSh 400; child KSh 100.

4. x+y=25, x–y=7; x=16, y=9.

5. Both equations are satisfied.

6. Substitution is useful when one variable is easy to isolate; elimination is useful when coefficients can be matched.

Circles: Circumference, Arc Length and Sector Perimeter

1. 66 cm.

2. 440 m.

3. 22 cm.

4. 22 2/3 cm.

5. 1 100 m = 1.1 km.

6. Radius centre-to-edge; diameter across centre; circumference distance around.

Area and Surface Area

1. 346.5 cm².

2. 20π cm².

3. 486 cm².

4. 376 cm².

5. 120π cm².

6. 150 cm².

7. Answers vary; estimate full squares and combine partial squares.

Interest, Appreciation, Depreciation and Hire Purchase

1. KSh 3 500.

2. KSh 18 675.

3. KSh 24 200.

4. KSh 68 000.

5. KSh 2 100 000.

6. KSh 38 000; extra KSh 3 000.

7. Simple interest uses original principal; compound interest includes accumulated interest.

Geometrical Constructions and Polygons

1. Construction drawing required.

2. Construction drawing required.

3. Construction drawing required.

4. 1 800°.

5. 135°.

6. 12 sides.

7. Construction drawing required.

8. Use an auxiliary ray, mark three equal intervals, join the last mark to the endpoint and draw parallels through the first two marks.

Coordinates and Graphs

1. I, II, III, IV respectively.

2. –4, –1, 2, 5, 8.

3. Straight line through listed points.

4. Answers depend on graph.

5. Intersection (2,4).

6. Consistent scale keeps distances proportional and points correctly positioned.

Scale Drawing

1. 2 km.

2. 1 cm represents 3 km; 1:300 000.

3. 6 cm by 4 cm.

4. 9.6 cm by 7.2 cm.

5. They represent large or small objects accurately on manageable paper.

6. 1:25 000.

Common Solids, Nets and Surface Area

1. Drawing required.

2. Rectangle $2\pi r$ by h plus two circles radius r.

3. 5 faces, 9 edges, 6 vertices.

4. 216 cm^2.

5. 297 cm^2 approximately.

6. 418 cm^2 approximately.

7. Unfold the solid into a flat net and measure the straight path on the net.

Data Presentation and Interpretation

1. Mean 7.75; median 8; mode 8.

2. Bar graph required.

3. Line graph required.

4. Bar graphs compare categories; line graphs show trends over continuous time/order.

5. Choose scale that covers data range and uses graph space clearly.

6. Mean uses all values but is affected by extremes; median resists extremes; mode identifies most common value.

Probability

1. 1/2; 3/10; 4/5.

2. 1/3.

3. 0.54 or 54%.

4. 7/20; 35%.

5. 1/4; 5/8.

6. Random variation, especially with a small number of trials.

FORMULA SUMMARY

Concept	Formula / rule
Integer subtraction	$a - b = a + (-b)$
Speed	Distance ÷ Time
Percentage change	(Change ÷ Original value) × 100%
Direct proportion	$y = kx$
Inverse proportion	$xy = k$
Simple interest	$I = PRT \div 100$
Compound amount	$A = P(1 + r/100)$■
Circumference	$C = \pi d = 2\pi r$
Arc length	$(\theta/360°) \times 2\pi r$
Sector perimeter	Arc length + 2r
Circle area	$A = \pi r^2$
Sector area	$(\theta/360°) \times \pi r^2$
Cube surface area	$6a^2$
Cuboid surface area	$2(lw + lh + wh)$
Cylinder curved surface area	$2\pi rh$
Closed cylinder total surface area	$2\pi r^2 + 2\pi rh$
Cone total surface area	$\pi rl + \pi r^2$
Polygon interior-angle sum	$(n - 2) \times 180°$
Each exterior angle of regular polygon	$360° \div n$
Mean	Sum of observations ÷ number of observations
Experimental probability	Frequency of event ÷ total trials
Theoretical probability	Favourable outcomes ÷ total equally likely outcomes